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Gender differences in interactions and depressive symptoms among hospitalized older patients living with dementia

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Abstract

Alzheimer's disease or a related dementia (ADRD) disproportionately affects women with two-thirds of individuals with ADRD comprised of women. This study examined gender-related differences in the quality of staff-patient interactions and depressive symptoms among hospitalized older patients living with dementia. This secondary analysis utilized baseline data of 140 hospitalized older patients with dementia who participated in the final cohort of a randomized controlled trial (ClinicalTrials.gov identifier: NCT03046121) implementing Family centered Function-focused Care (FamFFC). On average, the participants (male = 46.1%, female = 52.9%) were 81.43 years old (*SD* = 8.29), had positive interactions with staff and lower depressive symptoms based on Quality of Interaction Schedule (QUIS) scores and Cornell Scale for Depression in Dementia (CSDD) scores, respectively. Although males had more positive interactions (male = 6.06, *SD* = 1.13; female = 5.59, *SD* = 1.51) and lesser depressive symptoms (male = 7.52, *SD* = 4.77; female = 8.03, *SD* = 6.25) than females, no statistically significant gender differences were observed in linear models with appropriate covariates or multivariant analysis of covariant (MANCOVA). However, the multigroup regression conducted to further probe marginally significant moderation effect of gender and pain on staff-patient interactions demonstrated that greater pain was significantly related to lower quality or less positive staff-patient interactions for females compared to males (χ^2_{diff} (1) = 4.84, p = .03). Continued evaluation of gender differences is warranted to inform care delivery and interventions to improve care for hospitalized older patients with dementia.

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Keywords

Dementia; depressive symptoms; gender; hospitalized older adults; staff-patient interactions

Background

Alzheimer's disease or a related dementia (ADRD) currently affects an estimated 6.5 million Americans aged 65 and older and the prevalence is steadily increasing (Alzheimer's Association, 2022). ADRD disproportionately affects women with two-thirds of individuals with ADRD comprised of women (Alzheimer's Association, 2022; Kamalzadeh et al., 2019). The underlying reasons for higher lifetime risk of dementia among women are not completely clear and have included the longer life expectancy of women compared to men, hormonal differences between men and women, inflammation and metabolism, genetics, social roles and opportunities, and the interactions of all of these factors (Hasselgren et al., 2020; Nebel et al., 2018).

Gender and quality of interactions

Staff-patient interactions can be described as any verbal or nonverbal exchange between staff and patients and the quality of staff-patient interactions can be positive, negative, or neutral (Dean et al., 1993; McLean et al., 2017). Prior work has established that, in general, hospitalized persons with dementia or cognitive impairment tend to have more positive interactions compared to lessdesired negative and neutral interactions (Barker et al., 2016; Bridges et al., 2019; Lee et al., 2021). However, when present, negative encounters or lack of meaningful interactions are not only distressing to hospitalized patients but also associated with persons with dementia not receiving the same services as persons without dementia including attention to mobility, cognition, privacy, hygiene, and comfort (Moyle et al., 2011; Tolson et al., 1999). Thus, staff-patient interaction is a salient measure of quality of care in hospitalized patients with dementia.

There is limited work examining gender-related differences in the quality of staff-patient interactions. Furthermore, the existing studies predominantly focused on long-term care settings and reported mixed findings. For example, one prior study reported that male residents tend to have more interactions with staff than female residents (Lindesay & Skea, 1997). Another study reported that male residents have lower engagement and less personal attention from staff during daily care interactions compared to females (Shippee et al., 2015). In contrast, other research reports no significant association between gender and quality of interactions in settings of care (Bridges et al., 2019; McPherson et al., 2021). Understanding gender-related differences in the quality of staff-patient interactions among hospitalized patients with dementia is important because males and females have individual ways in which they interact, and it can be influenced by behavioral distress related to dementia. For example, on the one hand, females tend to be more polite, expressive, and emotional but less direct and more tentative during interactions; males, on the other hand, tend to be more focused, direct, and solution oriented (Basow & Rubenfeld, 2003; Palomares, 2009). Although female's polite and emotional interaction style could lead to higher quality or positive staff-patient interactions, the less-focused and tentative interaction coupled with behavioral distress such as paranoia, hallucination, delusion, or anxiety commonly noted in female patients with dementia (Colombo et al., 2018; Eikelboom et al., 2022) could lead to negative or neutral staff-patient interactions.

Gender and depressive symptoms

Depressive symptoms include feelings of negative emotions and sadness. Prior work has shown that older adults with dementia or cognitive impairment are likely to present with depressive symptoms such as social withdrawal, isolation, loss of interest in activities and hobbies, lack of energy, loss of appetite, self-depreciation, and pessimism (Alexopoulos et al., 1988; Leung et al., 2021; Yu et al., 2020). It has also been well established that presence of depressive symptoms have a significant negative impact in the quality of life of older adults, including those with dementia (Barbe et al., 2018; Lawton, 1994; Sivertsen et al., 2015).

Emerging research has provided evidence on the significant gender differences in depressive symptoms among older adults, including those with dementia or cognitive impairment (Hasselgren et al., 2020; Lövheim et al., 2009; Zhou et al., 2022). For example, female patients with dementia visiting the outpatient units in hospital settings were found to be more vulnerable to depressive symptoms compared to their male counterparts (Lee et al., 2017; Zhou et al., 2022). Another study including older adults with cognitive impairment in nursing homes, residential care units, and geriatric long-stay wards similarly reported that depressive symptoms are more common among women compared to men (Lövheim et al., 2009). A recent meta-analysis also reported that being female is associated with more prevalent and severe depressive symptoms (Eikelboom et al., 2022). While there is evidence of gender differences in depressive symptoms in persons with dementia in general, there is a lack of evidence that specifically focuses on hospitalized older patients with dementia.

The purpose of this study was to examine gender-related differences in the quality of staff-patient interactions and depressive symptoms among hospitalized older patients living with dementia. It was hypothesized that males will demonstrate more positive quality of staff-patient interactions and lesser depressive symptoms than females after accounting for possible covariates. Findings will advance our understanding of gender-related differences in the quality of staff-patient interactions and depressive symptoms among hospitalized older patients with dementia and support the need for greater attention to gender influences in future work involving older persons living with dementia. Understanding gender-related differences can also help to recognize the need for gender-specific interventions to improve the quality of staff-patient interactions and manage depressive symptoms among hospitalized older patients with dementia.

Methods

Study design and sample

This study was a secondary data analysis using baseline data of 140 hospitalized older patients with dementia who participated in the final cohort of a cluster randomized controlled trial (ClinicalTrials.gov identifier: NCT03046121). Data from the final cohort was included because this group of study participants were evaluated for quality of interactions, a measure not included in the prior two cohorts. The purpose of the parent study was to examine the efficacy of FamFFC (Family-centered Function-focused Care), a nurse-family caregiver partnership model that aims to improve: (a) the physical and cognitive recovery in hospitalized patients living with ADRD during hospitalization and the 60-day post-acute period; and (b) caregiver preparedness and experiences. The protocol has received institutional review board approval and has been published (Boltz et al., 2018).

The sample in this study was comprised of patients from two medical units of a large community teaching hospital in central Pennsylvania. Patients were eligible to participate if they were ≥65 years, spoke English or Spanish, lived in the community prior to the hospital admission, screened positive for dementia on scales with well-established psychometrics including the Montreal Cognitive Assessment, MoCA ≤ 25 (Bernstein et al., 2011; Gil et al., 2015; Nasreddine et al., 2005; Smith et al., 2007), the Eight-item Informant Interview to Differentiate Aging and Dementia, AD8 ≥ 2 (Galvin et al., 2005, 2006), and the Clinical Dementia Rating Scale, CDR = 0.5–2 (Morris, 1997), and had a designated family caregiver as a study partner. Patients were excluded if they had mild cognitive impairment (CDR = 0.5 without functional or ADL impairments), severe dementia (CDR = 3), any cognitive impairment associated neurological condition other than dementia (e.g., brain tumor), did not have a designated family caregiver as a study partner, were enrolled in hospice, were admitted from a nursing home, or experienced transfers to other unit for stays longer than 48 h.

Data collection and measures

Data were collected through a combination of chart review, interview of the patient/care partner with input from staff, and observation by trained research evaluators who had prior experience working with this population and were blinded to the intervention. Baseline data were used in this analysis, collected within 48 h of admission to the unit (Boltz et al., 2018).

The descriptive measures included patient's age, race/ethnicity, gender (identify as male/female), marital status, education, cognition, comorbidities, and length of stay. Data on descriptive measures were extracted through chart review except cognition which was evaluated using MoCA (Bernstein et al., 2011; Nasreddine et al., 2005). MoCA is a 10-min, 30-point cognitive screener designed to assist clinicians with detecting mild cognitive impairment among patients. It evaluates cognition through the domains of executive functioning, orientation, memory, abstract thinking, visuospatial abilities, language, and attention. A score of ≤25 on MoCA suggests mild cognitive impairment (Nasreddine et al., 2005; Smith et al., 2007). Prior research provided evidence of internal consistency (Cronbach's α = 0.79., 0.75), test-retest reliability (Intraclass Correlation Coefficient, ICC = 0.64–0.82), and inter-rater reliability (ICC = 0.87–0.99) of the MoCA (Bruijnen et al., 2020; Wong et al., 2018).

Other patient measures included the patient's level of function, pain status, provision of nonpharmacological measures related to physical well-being (e.g., listening to music, walking in the room/hall, validation, reminiscence, finger fidgets for exercise, audio/video tapes, warm blanket, hand/foot massage, and diversional activity), depressive symptoms, and quality of staff-patient interactions. The functional status of the patient was assessed using Barthel Index which includes 10 items related to self-care, mobility, and bowel/bladder functions (Mahoney & Barthel, 1965). The total scores range between 0 and 100 and higher scores suggest greater independence in functional activities. Prior work confirmed internal consistency of the Barthel Index and provided support for interrater reliability and construct validity (dos Santos Barros et al., 2022; Mahoney & Barthel, 1965; Sainsbury et al., 2005).

Patient's pain was assessed using the Pain Assessment in Advanced Dementia (PAINAD) Scale (Warden et al., 2003). PAINAD is an observational measure consisting of five domains —breathing, negative vocalizations, facial expression, body language, and consolability. Each of these five domains is scored 0–2 points. The total possible score is 10 and a score ≥2 indicates the presence of pain (Zwakhalen et al., 2012). Prior testing provided the evidence of internal reliability in persons with dementia (α = 0.90), strong inter-rater agreement (kappa = .74, $p < 0.000$) and concurrent validity (Kendall's τ = 0.73, $p < 0.000$) for the PAINAD (Mosele et al., 2012). Recent research also supported the validity and reliability of the PAINAD (α = 0.95 and infit/outfit statistics = 0.79–1.53) and reported no invariance of the PAINAD when used with black and white nursing home residents living with dementia (Resnick et al., 2021).

The assessment of depressive symptoms among the patients was done using Cornell Scale for Depression in Dementia (CSDD), a 19-item questionnaire which assesses the severity of depressive symptoms in those with dementia (Alexopoulos et al., 1988). The total scores in CSDD range from 0 to 38 and higher scores indicate more depressive symptoms. Prior research provided the evidence of internal consistency and supported predictive utility of the CSDD with sensitivity multivariant analysis of covariant = 0.83 and specificity = 0.73 for cutoff at ≥6, and sensitivity = 0.75 and specificity = 0.82 for cutoff at ≥8 (Williams & Marsh, 2009). More recent research also provided evidence of validity and reliability of the CSDD in Vietnamese and Persian population (Huynh et al., 2022; Mahmoudi et al., 2021).

The quality of staff-patient interactions was evaluated using the Quality of Interaction Schedule (QUIS). QUIS, an observational measure with five categories—positive social, positive care, neutral, negative protective, and negative restrictive—was developed to evaluate the quality of interactions in residential care units (Dean et al., 1993). It has been tested for psychometric properties and been used to evaluate quality of interactions between staff and older adults across settings of care (Dean et al., 1993; Jenkins & Allen, 1998; McLean et al., 2017; Mesa-Eguiagaray et al., 2016; Resnick et al., 2021). This study utilized the quantified QUIS where a research evaluator would observe a staff-patient interaction for about 20 min during any social or physical care activities and score each of the five QUIS categories from 0 to 2 (see Resnick et al., 2021). The total possible score ranged from 0 to 7 and higher scores indicated a better, or more positive, quality of interaction. Prior research provided the evidence of inter-rater reliability (Cohen's kappa multivariant analysis of covariant = 0.6–0.91) of the QUIS (Dean et al., 1993; Mesa-Eguiagaray et al., 2016) and additional testing using Rasch analysis supported internal consistency of the quantified QUIS with item separation of 6.33 and item reliability of .98 (Resnick et al., 2021).

The provision of nonpharmacological measures pertaining to diversion, physical activity, socialization, therapeutic communication, and comfort for patients was assessed using the investigator-developed Checklist of Non-Pharmacologic Intervention to Promote Well-Being. The checklist includes 31 potential interventions that are useful in promoting function and well-being of the patients. Data were collected through observations, report from staff, or chart extraction. It includes a binary response and items are checked "yes" if they were provided at any point during the patient's hospitalization. Thus, a patient can receive up to 31 non-pharmacological interventions during hospitalization.

Data analysis

Data analyses were completed using the Statistical Package for the Social Sciences (SPSS) version 28.0 (Arbuckle, 2022). First, we explored the descriptive characteristics of the participants to ascertain distributional characteristics and ensure that the assumptions associated with the planned statistical analyses were met. Then we performed *t*-tests (for continuous variables) and chi-square tests (for categorical variables) to report descriptive differences by gender. Following data check and descriptive reports, we conducted correlation analysis; patient's age, race, marital status, functional status, pain, and provision of nonpharmacologic measures were significantly associated with the dependent variables—quality of staff-patient interactions and depressive symptoms and thus used as covariates. We then proceeded for further analysis using multivariant analysis of covariant (MANCOVA) to examine gender differences as proposed in the study. MANCOVA is a type of multivariate analysis used to assess statistical differences on more than one dependent variable by an independent grouping variable while controlling for possible covariates. MANCOVA was used in this study to test the effect of the independent variable (gender) on a set of two dependent variables (quality of staff-patient interactions and depressive symptoms) with statistical and theoretical overlap while controlling for the identified covariates. All analyses considered a $p \leq .05$ level of significance.

Results

The sample included 140 patients, 66 males (46.1%) and 74 females (52.9%), with an overall mean age of 81.43 years (Standard Deviation, $SD = 8.29$). On average, the participants had some form of cognitive impairment based on a score of 6.47 ($SD = -.53$) in MoCA, stayed in the hospital for 7.30 days ($SD = 5.07$), had 5.65 comorbidities ($SD = 3.10$), and received 3.16 ($SD = 2.12$) non-pharmacological interventions. Additionally, the patients had moderate functional independence based on a mean Barthel Index score of 72.51 ($SD = 24.21$) and were observed to have none to mild pain based on a mean PAINAD score of 0.83 ($SD = 1.57$). Most of the participants were White/Caucasian ($n = 135$, 96.4%) and non-Hispanic/Latino ($n = 133$, 95%), half were married ($n = 71$, 50.7%), and majority had high school or higher level education ($n = 108$, 77.1%). On average, the participants in this study were engaged in positive interactions with staff based on a mean QUIS score of 5.84 ($SD = 1.36$) and had lower depressive symptoms based on a mean CSDD score of 7.79 ($SD = 5.59$).

As presented in Table 1, the independent *t*-test showed that hospitalized male and female patients with dementia in this study significantly differed in their levels of cognition (male = 15.36 ($SD = 5.56$), female = 12.47 ($SD = 6.94$); $t = 2.73$, $p = .007$) and quality of staff-patient interactions (male = 6.06 ($SD = 1.13$), female = 5.59 ($SD = 1.51$); $t = 2.05$, $p = .039$). Similarly, the chi-square test showed that hospitalized male and female patients significantly differed by marital status ($\chi^2 = 22.57$, $p < .001$).

As shown in Table 2, there were no statistically significant gender differences in the quality of staff-patient interactions and depressive symptoms observed in the multivariate tests based on MANCOVA ($F(2, 131) = 0.743$, $p = .477$). However, the multigroup multivariable regression conducted as an extended analysis to further probe the marginally significant moderation effect of gender and pain ($F(1, 126) = 3.675$, $p = .058$) on the quality of staff-patient interactions showed that greater pain is related to lower quality or less positive staff-patient interactions for females ($b = -0.324$, $p < .001$) compared to males, and this difference was significant ($\chi^2_{\text{diff}}(1) = 4.84$, $p = .03$). During preliminary testing for MANCOVA where we tested for any significant interaction between the independent variable (gender) and each covariate (patient's age, race, marital status, functional status, pain, and provision of nonpharmacologic measures), we noted a marginally significant interaction of gender and pain ($p = .058$) on the quality of staff-patient interactions suggesting a possible moderation effect. To probe this moderation effect, post MANCOVA, we followed up with a multigroup multivariable regression analysis in the Analysis of Moment Structures, AMOS version 28.0 (Arbuckle, 2022). The multigroup multivariable regression analysis allows simultaneous fitting of separate regression models for males and females to control for type 1 error while exploring group specific relationships. We tested whether the relationship between pain and quality of staff-patient interactions differed for males and females using nested model fit comparisons; model fit comparisons were determined using the change in chi-square (χ^2) and the standardized coefficient is provided as an indicator of effect size for individual estimate.

Discussion

The present study described the quality of staff-patient interactions and depressive symptoms among hospitalized older patients living with dementia and the gender-related differences. The hypothesis in this study was not supported in that we did not find significant gender differences in the quality of staff-patient interactions and depressive symptoms among hospitalized older patients with dementia after accounting for covariates. However, there was an extended finding of significant difference in the quality of staff-patient interactions between male and female patients with dementia moderated by pain in this study. For females, greater pain was associated with lower quality or less positive interactions with staff compared to males.

Our finding pertaining to gender differences in the quality of staff-patient interactions is in line with mixed findings from prior research where some studies reported significant gender differences (Lindesay & Skea, 1997; Shippee et al., 2015) and others did not find any associations between gender and quality of interactions (Bridges et al., 2019; McPherson et al., 2021) between staff and older adults, including those with cognitive impairment or dementia. The mixed results in existing research might mean that the findings are perhaps sample specific. Future work should continue to explore possible gender differences in the quality of staff-patient interactions, particularly among hospitalized older patients with dementia given the lack of evidence on gender differences in the quality of interactions in this population. Additionally, future work could also consider a comprehensive measurement of the quality of staff-patient interactions. A comprehensive measurement based on individual interaction skills (e.g., respectful listening and observation of the patient) as opposed to categories (e.g., positive, negative, or neutral) representative of the interaction skills might yield different findings.

We did not find significant gender differences in depressive symptoms in this study. This is in contrast to prior work which reported significant gender differences in depressive symptoms among older adults with dementia or cognitive impairment wherein being female was associated with more frequent and more severe depressive symptoms (Eikelboom et al., 2022; Hasselgren et al., 2020; Lee et al., 2017; Löovheim et al., 2009). The results, however, were in the desired direction with males having lower depressive symptoms than females based on lower mean CSDD scores for males. A possible reason for non-significant gender differences in depressive symptoms in this study could be the low levels of depression in our sample. Additionally, the sample included hospitalized older patients with dementia which is lacking in prior work. Future work should continue to explore the differences in depressive symptoms among hospitalized male and female older patients with dementia.

The extended finding of difference in the quality of staff-patient interactions between male and female patients with dementia moderated by pain in this study could be attributed to the difference in perception and expression of pain between males and females. There is some evidence suggesting that differences exist in the experience, processing, and expression of pain among males and females (Bartley & Fillingim, 2013; Mogil, 2020; Rhudy & Williams, 2005) including those with Alzheimer's disease or dementia (Cowan et al., 2017). For example, females seem to experience more chronic pain, report greater pain, and display comparatively higher sensitivity to pain while males, on the other hand, seem to have higher threshold and tolerance to pain than females (Eltumi & Tashani, 2017; Mogil, 2020). Staff-patient interaction is a mutual experience. It is likely that the greater pain among female patients with dementia interferes with their ability to interact in a friendly or positive manner with the staff resulting in neutral if not negative quality interactions. Likewise, expressions of underlying chronic pain, for example, with irritability or agitation might negatively impact the staff ability to interact with the females in a more desirable manner and result in lower quality or less positive interactions. Pain may also present as a behavioral symptom (e.g., resistance to care) and staff may respond negatively to the behaviors. Regardless, our findings indicate that staff looking to improve daily care interactions with patients with dementia in clinical settings need to be cognizant of pain among the patients, particularly female patients. Engagement in timely assessment and management of pain by providing appropriate pharmacological and/or non-pharmacological measures can support interactions and further help to improve patient experience and well-being in hospital settings. Similarly, the interventions to promote positive care interactions between staff and hospitalized patients with dementia should also educate staff on being cognizant of pain among the hospitalized patients with dementia with particular attention to females.

Limitations

Findings from this study are limited in that this secondary analysis utilized data from a study conducted in hospitals of one state thus making the findings more specific to the sample and less generalizable to the overall dementia population. In addition, our sample was small, primarily white and non-Hispanic. Future work can consider a more geographically and culturally diverse sample. The participants in this study also primarily interacted with female staff thus limiting our ability to examine differences in the quality of staff-patient interactions by staff gender. Whether staff gender impacts the quality of staff-patient interactions is an interesting area of investigation and should be considered in future dementia research. Additionally, the assessment of quality of staff-patient interactions was limited to rating the staff-patient interaction in five categories (i.e., positive, negative, or neutral) only and could be more comprehensive in the future work where the evaluation is based on individual interaction skills. Assessment of gender was also limited to identification as male or female and could be more comprehensive in the future work. Furthermore, findings may also be biased due to social desirability during observation of staff-patient interactions as well as recall bias on depressive symptoms questionnaire.

Conclusion

Despite some limitations, current study provides valuable information with useful implications and directions for future work with regard to the quality of staff-patient interactions and depressive symptoms among hospitalized male and female patients with dementia. The assessment of gender differences was based on an almost evenly split sample of male and female participants, and it provided some indication that there may be differences in the quality of staff-patient interactions based on whether the patient is male or female and their experience of pain. The extended finding of moderation effect of pain on gender and interactions is also valuable in that it informs the need to consider pain in both practice and research pertaining to interactions with hospitalized patients with dementia, particularly the need for greater attention to pain in hospitalized females with dementia. Continued evaluation of gender differences among hospitalized patients with dementia is needed to inform how care is provided to hospitalized male and female patients with dementia, and guide the design of interventions to improve care of hospitalized males and females living with dementia.

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Data availability statement

The data that support the findings of this study are available from the corresponding author, AP, upon reasonable request.

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Table 1.

Differences in patient characteristics by gender.

Characteristic	Male Mean (*SD*)/ *n* (%)	Female Mean (*SD*)/ *n* (%)	t/χ^2	*p*-Value
Age, years (range: 65–100)	81.95 (7.90)	80.96 (8.64)	0.708	.480
Cognition (MOCA, range: 0–25)	15.36 (5.56)	12.47 (6.94)	2.732	.007
Length of Hospital Stay, Days (range: 2–35)	7.79 (5.80)	6.86 (4.30)	1.064	.289
Comorbidities (CCI, range: 0–17)	6.00 (2.90)	5.23 (3.27)	1.222	.224
Functional Status (BI, range: 3–100)	73.92 (22.78)	71.26 (25.50)	0.649	.517
Non-pharmacological Interventions (range: 0–9)	3.00 (2.08)	3.30 (2.16)	−0.827	.409
Pain (PAINAD, range: 0–7)	0.61 (1.30)	1.03 (1.76)	−1.594	.113
Depressive Symptoms (CSDD, range: 0–32)	7.52 (4.78)	8.03 (6.25)	−0.540	.590
Quality of Staff-patient Interactions (QUIS, range: 2–7)	6.06 (1.13)	5.59 (1.51)	2.047	.039
Race			0.612	.434
White	65 (98.5)	70 (94.6)		
African American/Black	1 (1.5)	4 (5.4)		
Ethnicity			1.950	.162
Hispanic or Latino	1 (1.5)	6 (8.1)		
Not Hispanic or Latino	65 (98.5)	68 (91.9)		
Education			1.910	.167
Less than High School	39 (59.1)	53 (71.6)		
More than High School	27 (40.9)	21 (28.4)		
Marital Status			22.570	<.001
Not Married	18 (27.3)	51 (68.9)		
Married	48 (72.7)	23 (31.1)		

Note. $N = 140$ (Male = 66, Female = 74).

MoCA: Montreal Cognitive Assessment; CCI: Charlson Comorbidity Index; BI: Barthel Index; PAINAD: Pain Assessment in Advanced Dementia; CSDD: Cornell Scale for Depression in Dementia; QUIS: Quality of Interaction Schedule.

Table 2.

Multivariate analysis of outcomes by gender.

Independent variable	Dependent variable/s	*F*	*p*-Value	Partial Eta squared
Gender		0.743	.477	0.011
	Quality of Staff-patient Interactions, QUIS	1.198	.276	0.009
	Depressive Symptoms, CSDD	0.163	.687	0.001

Note. $N = 140$ (Male = 66, Female = 74). Covariates: patient's age, race, marital status, functional status, pain, and provision of nonpharmacologic measures. The Box's M was significant (Box's $M = 10.102$, $F = 3.314$, $p = .019$) so the Pillai-Bartlett trace was used to determine multivariate significance.

QUIS: Quality of Interaction Schedule: CSDD: Cornell Scale for Depression in Dementia.